

LOG-BOOK

WEEK-7:

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
from tensorflow.keras.callbacks import EarlyStopping

model = Sequential()
model.add(LSTM(92, input_shape=(X_train.shape[1], X_train.shape[2])))
model.add(Dense(1))

model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['mae']
)

model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
lstm_1 (LSTM)	(None, 92)	40,848
dense_1 (Dense)	(None, 1)	93

```
] from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
from tensorflow.keras.callbacks import EarlyStopping

model = Sequential()
model.add(LSTM(39, input_shape=(X_train.shape[1], X_train.shape[2])))
model.add(Dense(1))

model.compile(
    optimizer='adam',
    loss='mse',
    metrics=['mae']
)

model.summary()
```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
lstm_2 (LSTM)	(None, 39)	9,048
dense_2 (Dense)	(None, 1)	40

Total params: 9,088 (35.50 KB)

Trainable params: 9,088 (35.50 KB)

Non-trainable params: 0 (0.00 B)

```
] : early_stop = EarlyStopping(
    monitor='val_loss',
    patience=3,
    restore_best_weights=True
)

history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32,          # unchanged from practical
    epochs=10,
    batch_size=32,          # unchanged from practical
    validation_split=0.2,
    callbacks=[early_stop],
    verbose=1
)
```

```
Epoch 1/10
674/674 ————— 21s 24ms/step - loss: 0.0487 - mae: 0.0491 - val_loss: 4.4820e-04 - val_mae: 0.0192
Epoch 2/10
674/674 ————— 14s 20ms/step - loss: 1.8647e-04 - mae: 0.0111 - val_loss: 4.2256e-04 - val_mae: 0.0187
Epoch 3/10
674/674 ————— 14s 21ms/step - loss: 1.7511e-04 - mae: 0.0107 - val_loss: 4.0940e-04 - val_mae: 0.0182
Epoch 4/10
674/674 ————— 16s 23ms/step - loss: 1.5856e-04 - mae: 0.0102 - val_loss: 3.0731e-04 - val_mae: 0.0160
Epoch 5/10
674/674 ————— 13s 20ms/step - loss: 1.2440e-04 - mae: 0.0091 - val_loss: 2.1817e-04 - val_mae: 0.0129
Epoch 6/10
674/674 ————— 13s 19ms/step - loss: 7.5429e-05 - mae: 0.0070 - val_loss: 5.1263e-05 - val_mae: 0.0063
Epoch 7/10
674/674 ————— 13s 19ms/step - loss: 1.5620e-05 - mae: 0.0031 - val_loss: 8.2639e-06 - val_mae: 0.0024
Epoch 8/10
674/674 ————— 14s 21ms/step - loss: 9.6431e-06 - mae: 0.0024 - val_loss: 1.5458e-05 - val_mae: 0.0034
Epoch 9/10
674/674 ————— 13s 19ms/step - loss: 1.1728e-05 - mae: 0.0027 - val_loss: 7.8889e-06 - val_mae: 0.0023
Epoch 10/10
674/674 ————— 14s 21ms/step - loss: 1.0524e-05 - mae: 0.0026 - val_loss: 5.3816e-06 - val_mae: 0.0019
```

```
: test_mse, test_mae = model.evaluate(X_test, y_test, verbose=0)

print("Test MSE:", test_mse)
print("Test MAE:", test_mae)
```

Test MSE: 1.829516463658365e-06
Test MAE: 0.0010343455942347646

```
: import matplotlib.pyplot as plt

plt.figure()
plt.plot(history.history['mae'], label='Training MAE')
plt.plot(history.history['val_mae'], label='Validation MAE')
plt.xlabel('Epochs')
plt.ylabel('MAE')
plt.legend()
plt.show()
```

